Algebraic geometry 2026

Luiz Alberto

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Chapter 1

List of exercises

Exercise 1. Let $U = k^2 - \{0\}$, show that the set of regular functions $U \to k$ are polynomials, then conclude that U is not an affine variety.

Solution: Let $\varphi \in \mathcal{O}_U(U)$ be a regular function, then $\varphi|_{D_{k^2}(X)} = f/X^m$ and $\varphi|_{D_{k^2}(Y)} = g/Y^n$ and on the intersection $D_{k^2}(X) \cap D_{k^2}(Y) = D_{k^2}(XY)$, we have

$$Y^n f - X^m g = 0 \quad (*). \tag{1.1}$$

This means that $XY(Y^n f - X^m g) = 0 \iff Y^n f - X^m g = 0 \in k[X,Y]$. Now, let $f = \sum a_{ij} X^i Y^j$ and $g = \sum b_{ij} X^i Y^j$. Then, equality $(*)$ means

$$a_{i,j-n} = a_{i-m,j},\ \forall i \geq m, j \geq n. \tag{1.2}$$

Therefore $f/X^m = g/Y^n \in k[X,Y]$ and φ is a polynomial, because $U = D_{k^2}(X) \cup D_{k^2}(Y)$. In conclusion, we proved that the inclusion $i : U \hookrightarrow k^2$ induces a ring isomorphism

$$i^{\#}(k^2) : k[X,Y] \to \mathcal{O}_U(U) : f \mapsto f|_U \ (**). \tag{1.3}$$

Now assume $U = V(S),\ S \subset k[X,Y]$, then $f \in S \Rightarrow f|_U = 0$ and $f = 0$ because of $(**)$. So $S = \{0\}$ and $U = k^2$, which is a contradiction. □

Exercise 2. Let $V = V(XW - YZ) \subset \mathbb{A}^4_k$, such that $k[V] = k[X,Y,Z,W]/(XW - YZ)$. Let $\overline{X}, \overline{Y}, \overline{Z}, \overline{W}$ be the residue classes of X, Y, Z, W in $k[V]$. Then $\overline{X}/\overline{Y} = \overline{Z}/\overline{W} = f \in k(V)$ is defined at $P = (x,y,z,w) \in V$ if $y \neq 0$ or $w \neq 0$. Show that it is impossible to write $f = F/G$, with $F, G \in k[V]$ and $G(p) \neq 0$ for every point p where G is defined. Show that the set of poles of f is exactly $\{(x,y,z,w) \mid y = 0 \text{ and } w = 0\}$.

Solution: First we observe that $XW - YZ$ is a irreducible polynomial, because it's a degree one polynomial in $k[X,Y,Z][W] \subset \mathrm{Frac}(k[X,Y,Z])[W]$ and the Gauss lemma. In

particular $I(V) = (XW - YZ)$ is a prime ideal. Remember the isomorphism

$$k[V]_{(0)} \cong \mathrm{Frac}(k[V]) \to \mathcal{O}_{V,V} = k(V) : \frac{\overline{F}}{\overline{G}} \mapsto \langle D_V(G), \frac{F}{G} \rangle \quad (*). \tag{1.4}$$

Here, we write $\langle U, F/G \rangle$ for the equivalence class of $(U, F/G)$ in $\mathcal{O}_{X,X}$. so, by definition, we have $f = \langle D_V(Y), X/Y \rangle = \langle D_V(W), Z/W \rangle$. Now, write $f = \{(U_i, f_i) : i \in I\}$, then by definition the domain of f is the set $U = \bigcup_{i \in I} U_i$ and the set of poles of f is $P = V \setminus U$. First we show that $P = \{(x, 0, z, 0) \in V : x, z \in k\}$. Suppose that $f = \langle D_V(G), F/G \rangle$ (By isomorphisms $(*)$, any representative has this expression), then we have

$$YF - XG \in I(V) = (XW - YZ) \ (**). \tag{1.5}$$

Equation $(**)$ implies that $XG \in (Y, W)$, which is a prime ideal ($k[X, Y, Z, W]/(Y, W)$ is isomorphic to $k[X, Z]$, which is a domain). Then we see that $G \in (Y, W)$, because $X \notin (Y, W)$. In particular $D_V(G) \subset D_V(Y) \cup D_V(W)$ and this proves that $U = D_V(Y) \cup D_V(W)$, so $P = \{(x, 0, z, 0) \in k^4\}$ as we claimed. Furthermore, if there exists some representative $(D_V(G), F/G)$ of f such that $U = D_V(G)$, then $G|_{D_V(Y)} \in \mathcal{O}_V(D_V(Y))^*$, because $D_V(Y) \subset D_V(G)$.

Now we claim that $\mathcal{O}_V(D_V(Y))^* = k^* \langle Y^m, 1/Y^m : m \geq 0 \rangle$. In fact we have an isomorphism

$$k[V]_{\overline{Y}} \xrightarrow{\cong} \mathcal{O}_V(D_V(Y)) : \overline{F}/\overline{Y}^m \mapsto F/Y^m\big|_{D_V(Y)}. \tag{1.6}$$

Furthermore, we have another isomorphism

$$k[X, Y, W]_Y \to k[V]_{\overline{Y}} : F/Y^m \mapsto \overline{F}/\overline{Y}^m. \tag{1.7}$$

It's injective because $I(V) \cap k[X, Y, W] = (0)$ and it's surjective because we can use the relation $\overline{ZY} = \overline{XW}$ to write any element in $k[V]_{\overline{Y}}$ as a quotient $\overline{F}/\overline{Y}^m$, $F \in k[X, Y, W]$. However, $k[X, Y, W]^*_Y = k^* \langle Y^m : m \in \mathbb{Z} \rangle$ and we have proved the claim. Now, we see that there exist $m \geq 1, a \in k^*$ such that $G = aY^m$ on $D_V(Y)$, which means that $Y(G - aY^m)|_V = 0 \iff Y(G - aY^m) \in I(V) \iff G - aY^m \in I(V)$ because $Y \notin I(V)$. Therefore $G = aY^m$ on V and $D_V(G) = D_V(Y)$ which is a contradiction, because $(0, 0, 0, 1) \in D_V(W) \setminus D_V(Y)$. □

Exercise 3. Let X and Y be varieties, $P \in X$, and $Q \in Y$. Suppose that the local rings $\mathcal{O}_{X,P}$ and $\mathcal{O}_{Y,Q}$ are isomorphic (as k-algebras). Show that there exist open neighborhoods U of P in X and V of Q in Y such that U is isomorphic to V. Thus, the local ring of a variety at a point determines the variety, up to isomorphism, in a neighborhood of that point.

Solution: We can assume without lost of generality that $X \subset k^m$ and $Y \subset k^n$ are affine and irreducible. Write the isomorphism $\varphi : \mathcal{O}_{X,P} \to \mathcal{O}_{Y,Q}$ and write $\varphi(\langle X, x_i \rangle) = \langle V_i, g_i \rangle$, where $x_i = X_i|_X$ is the restriction of the ith coordinate function of k^m to X. Since φ is an isomorphism, we have $\varphi(\mathfrak{m}_{X,P}) = \mathfrak{m}_{Y,Q}$, so φ descends to an isomorphism $\overline{\varphi} : \mathcal{O}_{X,P}/\mathfrak{m}_{X,P} \to \mathcal{O}_{Y,Q}/\mathfrak{m}_{Y,Q}$. So we have the diagram

$$\begin{array}{ccc} \mathcal{O}_{X,P}/\mathfrak{m}_{X,P} & \xrightarrow{\overline{\varphi}} & \mathcal{O}_{Y,Q}/\mathfrak{m}_{Y,Q} \\ {\scriptstyle \mathrm{ev}_P}\downarrow & & \downarrow{\scriptstyle \mathrm{ev}_Q} \\ k & \dashrightarrow & k \end{array}$$

where the dashed line is a k-algebra isomorphism of fields, which must be the identity map. Then we see that $\mathrm{ev}_Q \circ \overline{\varphi} = \mathrm{ev}_P$. Now define the morphism

$$g : V = \bigcap_i V_i \to k^m : y \mapsto (g_1(y), \ldots, g_n(y)). \tag{1.8}$$

We claim that $g(V_0) \subset X$ for some neighborhood V_0 of Q. In fact, if $F \in I(X)$ we have

$$\langle V, F(g_1, \ldots, g_m)\rangle = \varphi(\langle X, F(x_1, \ldots, x_m)\rangle). \tag{1.9}$$

But $F(x_1, \ldots, x_m) = F(X_1, \ldots, X_m)|_X = 0$, so $F(g_1, \ldots, g_m) = 0$ on some neighborhood $V_0 \ni Q, V_0 \subset V$. Therefore $F(g_1(y), \ldots, g_m(y)) = 0$ and $g(y) \in VI(X) = X$ for $y \in V_0$. Furthermore, we have $g_i(Q) = \mathrm{ev}_Q(g_i) = \mathrm{ev}_Q \circ \overline{\varphi}(x_i) = \mathrm{ev}_P(x_i) = x_i(P)$, so $g(Q) = P$. For simplicity of notation we will keep writing $V_0 = V$. Then we see that we constructed a morphism $g : V \to X$ which maps $Q \mapsto P$. In a similar way we construct a morphism $f : U \to Y$ which maps $P \mapsto Q$. Now we compose

$$V \cap g^{-1}(U) \xrightarrow{g} U \xrightarrow{f} X \ (*). \tag{1.10}$$

We claim that this composition is the inclusion. We have

$$\varphi(\langle X, x_i\rangle) = \langle V, g_i\rangle,\ \varphi^{-1}(\langle Y, y_i\rangle) = \langle U, f_i\rangle. \tag{1.11}$$

Let $f_i = a_i/b_i$, where $a_i = A_i|_X, b_i = B_i|_X$ for $A_i, B_i \in k[\underline{X}]$. Then we get

$$\begin{aligned} \varphi\varphi^{-1}(\langle Y, y_i\rangle) &= \varphi(\langle U, f_i\rangle) \\ &= \varphi(\langle U, \frac{A_i(x_1, \ldots, x_m)}{B_i(x_1, \ldots, x_m)}\rangle\rangle) \\ &= \varphi(\frac{A_i(\langle U, x_1\rangle, \ldots, \langle U, x_m\rangle)}{B_i(\langle U, x_1\rangle, \ldots, \langle U, x_m\rangle)}) \\ &= \frac{A_i(\langle V, g_1\rangle, \ldots, \langle V, g_m\rangle)}{B_i(\langle V, g_1\rangle, \ldots, \langle V, g_m\rangle)} \\ &= \langle V, \frac{A_i(g_1, \ldots, g_m)}{B_i(g_1, \ldots, g_m)}\rangle. \end{aligned} \tag{1.12}$$

So $\frac{A_i(g_1,\ldots,g_m)}{B_i(g_1,\ldots,g_m)} = y_i$ on some neighborhood of $V' \ni Q$ in V. This proves that the composition $(*)$ is the inclusion $V' \hookrightarrow X$. This prove that there exists open sets $V \subset Y, U \subset X$ such that $f \circ g = \mathbb{1}_V, g \circ f = \mathbb{1}_U$, so $g : V \to U$ is an isomorphism. □

Exercise 4. Let X be a variety and $z \in k(X)$.
(a) Show that the set of poles of z is a closed subset of X.
(b) If $z \in \mathcal{O}_{X,P}$, show that there exists an open neighborhood U of P such that $z \in \mathcal{O}(U)$, that is, z is regular on U.
(c) Conclude that $\mathcal{O}_{X,P} = \bigcup_U \mathcal{O}(U)$, where the union is taken over all open neighborhoods U of P in X.

Solution: First, we have $z = \{(U_i, f_i) : i \in I\}$ where $f_i \in \mathcal{O}(U_i)$. The set of poles is by definitions $P = X \setminus U, U = \bigcup_i U_i$, so it's closed. Now, by definition if $(U, f) \in z$, then $f \in \mathcal{O}(U), U \ni p$, so z is in the image of the embedding $\mathcal{O}(U) \hookrightarrow \mathcal{O}_{X,P}$. This also proves that $\mathcal{O}_{X,P} = \bigcup_{U \ni P} \mathcal{O}(U)$. □

Exercise 5. Let X be a projective variety and $P, Q \in X$. Show that there exists an affine open subset of X containing P and Q.

Solution: We assume $X = \mathbb{P}^m$ and let $P, Q \in k^{m+1}$ be linearly independent. Then, there exists $H \in (k^{m+1})^*$ such that $H(P) = H(Q) \neq 0$ and $P, Q \in X \setminus V(H)$. We claim that $X \setminus V(H) \cong k^m$. To see this choose a basis of k^{m+1} and let $H = (a_1, \ldots, a_m)$ be the associated matrix representation of H. Then if $T \in GL_{m+1}(k)$, its descends to an isomorphism of X and $T^{-1}(V(H)) = V(H \circ T)$. This means that whenever we do a change of coordinates on k^{m+1}, we do an elementary operation with the collumns of H. Since $H \neq 0$, there exists a change of coordinates T such that $H \circ T = (0, \ldots, 0, 1) = X_{m+1}$. Therefore $T^{-1}V(H) = V(X_{m+1})$ and $T^{-1}(X \setminus V(H)) = X \setminus V(X_{m+1}) \cong k^{m+1}$. □

1.1 Vector bundle list

Definition. Let X be a variety. In the exercises below, E is always a closed subset of $X \times \mathbb{A}^N$, for some N, and $p\colon E \to X$ is the restriction of the projection onto the first factor. We say that $p\colon E \to X$ is a (algebraic) vector bundle of rank r if there exist an open cover $X = \bigcup_i U_i$ and isomorphisms

$$\phi_i\colon p^{-1}(U_i) \to U_i \times \mathbb{A}^r$$

such that:

(i) ϕ_i is compatible with the projections, that is, $\phi_i(e) = (p(e), *)$ for all $e \in p^{-1}(U_i)$;

(ii) for each $P \in U_i$, the restriction of ϕ_i to the fiber $p^{-1}(P)$ is a linear isomorphism onto $\{P\} \times \mathbb{A}^r = k^r$.

In this case, on $U_i \cap U_j$ we have $\phi_i \circ \phi_j^{-1}(P, w) = (P, g_{ij}(P)w)$, where $g_{ij}\colon U_i \cap U_j \to \mathrm{GL}_r(k)$ is a morphism; the g_{ij} are the transition functions. A section of E is a morphism $s\colon X \to E$ with $p \circ s = \mathrm{id}_X$.

In the solution of the exercises, i will use p_1 for the projection on the first coordinate and p_2 for the projection on the second coordinate whenever it's necessary.

Exercise 1 (Tangent bundle of a hypersurface). Let $F \in k[X_1, \ldots, X_n]$ be an irreducible polynomial and let $V = V(F) \subset \mathbb{A}^n$. Assume that V is nonsingular, that is, that for every $P \in V$ some partial derivative $F_{X_j}(P)$ is nonzero. Let

$$TV = \Big\{(P, v) \in V \times \mathbb{A}^n \;\Big|\; \sum_{i=1}^{n} F_{X_i}(P)\, v_i = 0\Big\}.$$

(a) Show that TV is a closed subset of $V \times \mathbb{A}^n$ and that, for each $P \in V$, the fiber $p^{-1}(P)$ is the tangent space $T_P(V)$ (see Exercise 3.11 of Fulton).

Solution: First we remember the isomorphism

$$\{v \in k^n : \sum_i D_i(F)|_p v_i = 0\} \xrightarrow{\cong} T_pV = \mathrm{Der}_k(k[V], k_p) \tag{1.13}$$

which sends a vector v to the corresponding derivation $D_v : f \mapsto \sum_i D_i(f)|_p v_i$. From this we see that $TV|_p \cong T_pV$. Moreover $TV \subset V \times k^n$ is closed. In fact, we have

$$\begin{aligned} TV &= \{(p, v) \in V \times k^n : \sum_i p_1^*(D_iF) p_2^*(X_i)(p, v) = 0\} \\ &= V(\sum_i p_1^*(D_iF) p_2^*(X_i)) \cap (V \times k^n) \end{aligned} \tag{1.14}$$

□

(b) For $j = 1, \ldots, n$, let $U_j = \{P \in V \mid F_{X_j}(P) \neq 0\}$. Show that the U_j form an open cover of V and that

$$p^{-1}(U_j) \to U_j \times \mathbb{A}^{n-1}, \qquad (P, v) \mapsto (P, (v_1, \ldots, \widehat{v_j}, \ldots, v_n))$$

is a linear isomorphism on the fibers. Conclude that TV is a vector bundle of rank $n - 1$ over V.

Solution: We have $U_j = D_{TV}(p_1^*(D_jF))$ which forms an open covering because V is nonsingular. Moreover, $p_1^{-1}(U_j) \to U_j \times k^n : (p, v) \mapsto (p, v_1, \ldots, \hat{v}_j, \ldots, v_n)$ is a morphism, such that

$$E|_p \to \{p\} \times k^n \tag{1.15}$$

is an isomorphism, because $v_j = \frac{1}{D_jF|_p} \sum_{i \neq j} D_i(F)|_p v_i$. □

(c) Assume $\mathrm{char}\, k \neq 2$ and let $C = V(X^2 + Y^2 - 1) \subset \mathbb{A}^2$. Show that C is nonsingular and that

$$C \times \mathbb{A}^1 \to TC, \qquad ((x, y), \lambda) \mapsto ((x, y), \lambda(-y, x))$$

is an isomorphism, linear on the fibers. Conclude that the tangent bundle of C is trivial.

Solution: We have $C = V(F) \subset k^2,\ F = X^2 + Y^2 - 1$, so

$$(\frac{\partial F}{\partial X}(p), \frac{\partial F}{\partial Y}(p)) = 0 \iff 2X(p) = 2Y(p) = 0 \iff p = 0. \tag{1.16}$$

Since $p \neq C$, we see that C is nonsingular. Moreover

$$C \times k^1 \to TC : ((x, y), \lambda) \mapsto ((x, y), \lambda(-y, x)) \tag{1.17}$$

is a morphism becasue $\lambda(\frac{\partial F}{\partial X}(x, y)(-y) + \frac{\partial F}{\partial Y}(x, y)x) = 0$. Furthermore, on each fiber we have $\{p\} \times k^1 \to TC|_p : \lambda \mapsto \lambda(-y, x)$, which is an isomorphism because $(-y, x) \neq 0$. □

Exercise 2. Let

$$G = V(p_{12}p_{34} - p_{13}p_{24} + p_{14}p_{23}) \subset \mathbb{P}^5.$$

Recall that the subspace Λ generated by the rows of a 2×4 matrix A of rank 2 corresponds to the point $[p_{ij}]$ of the 2×2 minors of A (columns $i < j$). Let

$$S = \{(\Lambda, v) \in G \times \mathbb{A}^4 \mid v \in \Lambda\}.$$

(a) Show that S is the closed subset of $G \times \mathbb{A}^4$ defined by the equations

$$p_{ij}v_i - p_{il}v_j + p_{ij}v_l = 0, \qquad 1 \leq i < j < l \leq 4.$$

Hint: $v \in \Lambda$ if and only if the 3×4 matrix obtained by adding v as the third row of A has rank ≤ 2.

Solution: Let $\Lambda = A \in \mathrm{Mat}_{2\times 4}(k)$, so $v = (v_1, v_2, v_3, v_4) \in \Lambda$ if and only if

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ v_1 & v_2 & v_3 & v_4 \end{pmatrix} \tag{1.18}$$

has rank 2. This turns out to be equivalent to

$$v_i \begin{vmatrix} a_{1j} & a_{1l} \\ a_{2j} & a_{2l} \end{vmatrix} - v_j \begin{vmatrix} a_{1i} & a_{1l} \\ a_{2i} & a_{2l} \end{vmatrix} + v_l \begin{vmatrix} a_{1i} & a_{1j} \\ a_{2i} & a_{2j} \end{vmatrix} = 0. \tag{1.19}$$

□

(b) Let $U_{12} = G \cap \{p_{12} \neq 0\}$. Show that every $\Lambda \in U_{12}$ is generated by the rows of a unique matrix of the form

$$A_\Lambda = \begin{pmatrix} 1 & 0 & a & b \\ 0 & 1 & c & d \end{pmatrix},$$

and that $\Lambda \mapsto (a, b, c, d)$ is an isomorphism $U_{12} \cong \mathbb{A}^4$. Compute the Plücker coordinates of Λ in terms of a, b, c, d.

Solution: Through the Plücker embedding, we have $\Lambda \to \mathbb{P}^5 : A_\Lambda \mapsto (1 : c : d : -a : -b : ad - bc)$ which is mapped to $(c, d-a, -b, ad-bc)$. Then we have an isomorphism

$$U_{12} \xrightarrow{\cong} \{(c, d, -a, -b, ad - bc) \in k^5\} \tag{1.20}$$

which is the graph of the morphism $(c, d, a, b) \mapsto bc - da$. This graph is isomorphic to k^4 through the projection on the first four coordinates. □

(c) Show that

$$p^{-1}(U_{12}) \longrightarrow U_{12} \times \mathbb{A}^2, \qquad (\Lambda, v) \longmapsto (\Lambda, (v_1, v_2))$$

is a linear isomorphism on the fibers. Using the analogous open sets $U_{ij} = G \cap \{p_{ij} \neq 0\}$, conclude that S is a vector bundle of rank 2 over G, called the tautological bundle of G.

Solution: We have that $p^{-1}(U_{12}) \to U_{12} \times k^2 : (\Lambda, v) \mapsto (\Lambda, v_1, v_2)$ is a morphism of algebraic varieties. Furthermore, the restriction on fibers give an isomorphism of vector spaces. In fact we have $S|_\Lambda \to \{\Lambda\} \times k^2 : v \mapsto (v_1, v_2)$ is an isomorphism because of the relations

$$\begin{cases} v_1(-a) - v_2 c + v_3 = 0 \\ v_1(-b) - v_2(d) + v_4 = 0 \end{cases} \tag{1.21}$$

Here we are takeing a representative A_Λ as in the item (b). □

(d) Let $U_{13} = G \cap \{p_{13} \neq 0\}$, with the analogous trivialization $(\Lambda, v) \mapsto (\Lambda, (v_1, v_3))$. Determine the transition function $g \colon U_{12} \cap U_{13} \to \mathrm{GL}_2(k)$ between the two trivializations and show that $\det g = p_{13}/p_{12}$. Compare with the transition function of the tautological line bundle of $\mathbb{P}^5$ between the charts $\{p_{12} \neq 0\}$ and $\{p_{13} \neq 0\}$.

Solution: The transition matrix $g : U_{12} \cap U_{13} \to GL_2(k)$ is such that

$$U_{12} \cap U_{13} \times k^2 \to U_{12} \cap U_{13} \times k^2 : (\Lambda, v_1, v_2) \mapsto (\Lambda, g \cdot (v_1, v_2)) = (\Lambda, v_1, v_3). \tag{1.22}$$

However, by item (a), we have

$$v_1 \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} - v_2 \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} + v_3 \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = 0, \tag{1.23}$$

i.e., we have $v_3 = \frac{p_{13}v_2 - p_{23}v_1}{p_{12}}$. So the transition matrix is $g = \begin{pmatrix} 1 & -\frac{p_{23}}{p_{12}} \\ 0 & \frac{p_{13}}{p_{12}} \end{pmatrix}$. Its determinant is $\det(g) = \frac{p_{13}}{p_{12}}$.

The tautological line bundle of $\mathbb{P}^5$, is defined by

$$\mathcal{O}_{\mathbb{P}^5}(-1) = \{(L, v) \in \mathbb{P}^5 \times k^6 : v \in L\}. \tag{1.24}$$

If $L = (a_1 : \cdots : a_6)$ and $v = (v_1, \ldots, v_6)$, then $v \in L$ if and only if

$$\begin{pmatrix} a_1 & a_2 & a_3 & a_4 & a_5 & a_6 \\ v_1 & v_2 & v_3 & v_4 & v_5 & v_6 \end{pmatrix} \tag{1.25}$$

has rank 2. So each 2×2-minor has determinant zero, which is equivalent to $a_j v_i - a_i v_j = 0$. These relations proves that $\mathcal{O}_{\mathbb{P}^5}(-1)|_{U_j} \to U_j \times k^1 : (L, v) \mapsto (L, v_j)$ gives local trivializations of $\mathcal{O}_{\mathbb{P}^5}(-1)$. Finally, the relation $v_3 = \frac{a_3}{a_2} v_2$ gives the transition function $g = \frac{p_2}{p_3}$ from U_2 to U_3. □

(e) Show that the only section of S is the zero section.

Solution: A section of S induces a section of $G \times k^2$:

$$\begin{array}{ccccc} G & \longrightarrow & S & \hookrightarrow & G \times k^2 \\ & \searrow^{\mathbb{1}_G} & \downarrow & \swarrow & \\ & & G & & \end{array}$$

Howver, a section of the trivial bundle $G \times k^2$ is a regular function $G \to k^2$, which must be a constant v. However, by definition of the section we have $v \in \Lambda$ for any $2-$dimensional subspace $\Lambda \subset k^4$. Therefore $v = 0$. □